Kendall's W

Viva questions and answers | a plain-language guide to the method, project results, limitations and next steps

The short answer to memorize

Kendall's W measures whether the same months tend to rank high or low across different years. In my project, I ranked the 12 calendar months by outbreak-event count separately for 2022, 2023, 2024 and 2025, then measured how much the four yearly rankings agreed. ADD had the clearest agreement (W = 0.530, p = 0.016). I describe this as exploratory because it is based on only four years and the p-value uses an approximation.

What does the "12 months" part mean?

The 12 months are the things being ranked: January through December. For each completed year, we count how many outbreak events were reported in each month and rank the months from busier to quieter. We then compare those four rankings. The question is whether the relative monthly pattern repeats, not whether every year had the same number of events.

Year	Jan	Feb	Mar	...	Oct	Nov	Dec
2022	Monthly event counts are converted into a ranking						
2023	Monthly event counts are converted into a ranking						
2024	Monthly event counts are converted into a ranking						
2025	Monthly event counts are converted into a ranking						

Imagine four people each rank the same 12 months from busiest to quietest. If their lists look alike, agreement is high. If each year has a very different order, agreement is low. Kendall's W summarizes that overall agreement in one number.

How to read W

W ranges from 0 to 1. A value closer to 0 means little agreement among yearly month rankings; a value closer to 1 means strong agreement. W is not a percentage, does not identify a cause, and does not mean month counts were equal. It is an overall measure of rank agreement.

How Kendall's W was used

The baseline contains four complete calendar years, 2022-2025. For each disease family, the analysis compared that disease's 12 monthly outbreak-event counts across those years. This tests whether a disease's busier and quieter months tend to keep a similar relative ordering year to year.

Disease	W	p-value	Plain-language reading
Acute Diarrheal Disease (ADD)	0.530	0.016*	Most agreement; only result below 0.05 in the report
Dengue	0.405	0.086	Some agreement, not significant at 0.05
Food Poisoning	0.386	0.108	Some agreement, not significant at 0.05
Malaria	0.319	0.229	Lower agreement; not significant
Chikungunya	0.150	0.832	Very little agreement in this test

* ADD also remains significant after the report's false-discovery-rate adjustment (q = 0.034), but not under the stricter Bonferroni threshold. The report calls the finding exploratory because there are only four yearly profiles and the p-value relies on an approximate method.

Q: What does W = 0.530 for ADD tell us?

A: It indicates a meaningful level of agreement in ADD's month rankings across 2022-2025. The p-value of 0.016 is below the conventional 0.05 threshold in the report, so ADD is the only disease with statistically significant concordance in this test. I still call it exploratory because only four yearly profiles were compared.

Q: What does p = 0.016 mean?

A: If there were actually no year-to-year rank agreement, a result at least this extreme would be relatively unlikely under the test's assumptions. It is not a 1.6% chance that the finding is wrong, and it does not prove the pattern will continue in the future.

Q: Does a non-significant result mean there is no seasonality?

A: No. Kendall's W asks whether the detailed ranking of months repeats across years. A disease can be seasonally concentrated overall while its exact busiest month changes. Dengue is an example: pooled timing is concentrated in June-October, but the precise peak shifts between years, so its W result is not significant.

Likely viva questions

Q: Why did you use Kendall's W?

A: My question was about agreement among several yearly rankings. Kendall's W summarizes concordance among multiple rankings and works with ranks rather than requiring monthly counts to follow a normal distribution. It fits this short four-year comparison better than claiming a long-term forecasting model.

Q: Why are there 12 months and only four years?

A: Each calendar month is ranked within each year, so there are 12 items being ranked. The four complete years are the four separate rankings being compared. The incomplete 2026 period is kept out of this baseline because it does not cover a full year.

Q: Are you comparing cases or outbreak events?

A: Kendall's W compares monthly outbreak-event counts. An event is a notified outbreak cluster; cases are the number of people recorded within those clusters. A few large clusters can dominate case totals, so event frequency and case volume answer different questions.

Q: What are the main limitations of this result?

A: There are only four annual profiles, so the result is exploratory and the p-value approximation may be imperfect. The dataset records notified outbreaks, not every illness in the population. Reporting differs between places and years, and the analysis does not include weather measurements or establish causes. The result should not be treated as a forecast.

Q: Why not include 2026?

A: The report has 2026 only through epidemiological week 32. It is an incomplete year, so including it as though it were a full calendar year would make the monthly profile incomplete and not directly comparable. The report treats it separately as a partial-year check.

Q: Does Kendall's W show that rainfall causes disease?

A: No. It measures similarity in month rankings across years. No weather dataset was linked in this analysis, so the results show timing and repeatability in reported outbreak data, not climate causation.

Q: What did you learn from this analysis?

A: I learned to separate a visible seasonal pattern from a pattern that repeats reliably every year. I also learned to interpret a p-value carefully, distinguish outbreak events from cases, and describe the limits of surveillance data instead of making claims the data cannot support.

Q: What would you do next?

A: I would add more complete years, evaluate the Kendall result with an exact or permutation-based method suited to a small number of years, and link records to verified weather and population data if available. I would also compare reporting coverage across districts and validate the patterns with more field sites.

A confident 30-second explanation

I used Kendall's W to ask whether the same months tend to be relatively busier or quieter from one year to another. For each disease, I ranked the 12 months by outbreak-event counts separately for 2022, 2023, 2024 and 2025, then measured agreement among those four yearly rankings. ADD had the strongest agreement, W = 0.530, with p = 0.016. It was the only disease significant at the 0.05 level in the report. Because the analysis has only four years and uses an approximate p-value, I describe the result as exploratory. It measures repeatability, not disease incidence or climate causation.

Three points to keep straight

Term	What it means
Kendall's W	How much the yearly month rankings agree. It ranges from 0 to 1.
p-value	How unusual the observed result would be under the no-agreement assumption, subject to the test assumptions.
Seasonality	Months may be concentrated in one part of the year even when the exact month order changes between years.

If the examiner challenges the result

Say: "The data support a repeatable monthly ranking for ADD in this four-year baseline, but I am careful not to overstate it. The report labels it exploratory because the number of years is small. More years and a small-sample inference check would strengthen the conclusion."

Source note

Prepared from the finalized report: Kendall's W definition (printed report p. 14), stability results and interpretation (pp. 32-33), and viva preparation guide (p. 54). Use the finalized report as the authority if an older project note gives different figures.